



UK Research
and Innovation

iDeer Tool

Technical Document

Version and updates

Document version	iDeer Tool version	Date published	Update details
1.0	1.0	05/12/2025	-

This technical documentation accompanies the iDeer Tool version 1.0. The models used to make predictions are based on 2023 versions of spatial data inputs (e.g., land cover data). To maintain the relevance of outputs, where possible the iDeer Tool will be updated as new spatial data become available. Any changes to this document or the iDeer Tool will be outlined in the table above.

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What is the iDeer Tool?

The iDeer Tool is an interactive decision support tool for **England and Wales** that predicts and maps Deer Impact Risk to woodlands at a landscape scale. The iDeer Tool predicts impact risk for two groups of deer species: large deer (fallow, sika and red deer) and small deer (roe, muntjac and Chinese water deer). Users of the iDeer Tool can view the current risk of deer impacts to all woodland parcels across an area of interest to them, then visualise how the distribution and severity of Deer Impact Risk might change under different woodland creation scenarios. Such information can inform and guide land use decisions and deer impact mitigation efforts on the ground.

❖ To access the tool, click on this [weblink](#)

Why is the iDeer Tool needed?

Tree planting and woodland management, informed by scientific evidence, are essential nature-based solutions to tackling the biodiversity and climate crises. Whilst deer can have important benefits for nature and people, large deer populations and shifting species ranges represent a major barrier to meeting woodland expansion targets and management objectives. Combining successful woodland expansion schemes with effective deer management plans is therefore critical for meeting net zero and biodiversity targets.

Deer management is costly and complex. Several methods are used to mitigate unwanted deer impacts, including shooting and fencing. Shooting requires a firearms license and training with rifles of sufficient calibre, while fencing is expensive and requires regular maintenance. Both of these methods can be highly effective, but require considerable investment of time and money. Deer are highly mobile and cross jurisdictional boundaries, and land ownership in England and Wales is typically highly spatially divided. Therefore, whilst one landowner may invest considerable resources into deer management, they may receive minimal benefit if their neighbours do not also participate. In addition, decisions made by one landowner, such as planting highly nutritious saplings, can influence deer impacts on their neighbours if deer behaviour is affected. To minimise conflict between neighbours and achieve successful woodland and deer management, co-ordination at the landscape-scale is essential. However, identifying the current risk of deer impacts and anticipating the potential changes to impact risk following woodland creation can be challenging, even for experienced practitioners.

Here we present the iDeer Tool, an interactive decision support tool for **England and Wales** that predicts and maps Deer Impact Risk to woodlands at a landscape scale. The tool can be used to: 1) View the current Deer Impact Risk for all woodland parcels across a landscape, 2) Design 'future landscapes', comprised of alternative woodland parcel creation scenarios and 3) Assess how different woodland creation scenarios might influence the risk of deer impacts to woodland parcels in the future.

Who is the iDeer Tool for?

The iDeer Tool is designed to be used by **anyone involved in woodland or land management**, or anyone who wants to understand more about potential deer impacts to new and existing woodlands. Users of the iDeer Tool do not require specialist knowledge or training. Instructions for using the iDeer Tool and interpreting the maps that it produces are provided in the Instruction tab of the iDeer Tool.

The iDeer Tool has been designed to support land-use decisions of anyone involved in land management. This includes individuals, **local authorities** and **environmental organisations** involved in woodland management, woodland creation, or the management of deer.

The iDeer tool provides downloadable Deer Impact Risk maps which can be included in Deer Management Plans and other reports when applying for grants or government funding.

How is Deer Impact Risk calculated?

Deer cause impacts to woodland through browsing foliage and removing the bark of trees and grazing and trampling woodland ground vegetation. We define **Deer Impact Risk** as the likelihood that a woodland will experience any of these types of deer-related damage. Our estimates of this likelihood are based on predicted deer activity levels in woodlands, given current understanding of deer behaviour.

The Deer Impact Risk score is calculated using a Bayesian Belief Network (BBN) model, which provides information on the variables determining Deer Impact Risk (e.g., woodland characteristics)

based on expert opinion. **Deer Impact Risk is presented as a five-point score ranging from 1 (low risk) to 5 (high risk).**

The model was iteratively co-developed with an expert panel comprising deer managers and scientists (**Figure 1**). Consultation with published scientific research as well as several rounds of expert consultation and validation were carried out, including a workshop, questionnaires, and semi-structured interviews, to identify key variables to include in the model, and define how they interact to influence impact risk.

Experts identified that the deer species in England and Wales could be split into two groups for predicting impact risk. Large-bodied species (red deer, Japanese sika deer and fallow deer) range over larger areas and tend to graze on grasses and other low vegetation, whereas smaller-bodied species (roe deer, Chinese water deer, Reeve's muntjac) have more localised movement within territories and mostly browse on trees and shrubs. Therefore, we produced two separate models for large deer species and small deer species and predicted impact risk separately for these groups. Risk maps are presented separately for both groups in the tool. The model structures are presented in influence diagrams in **Figure 2**.

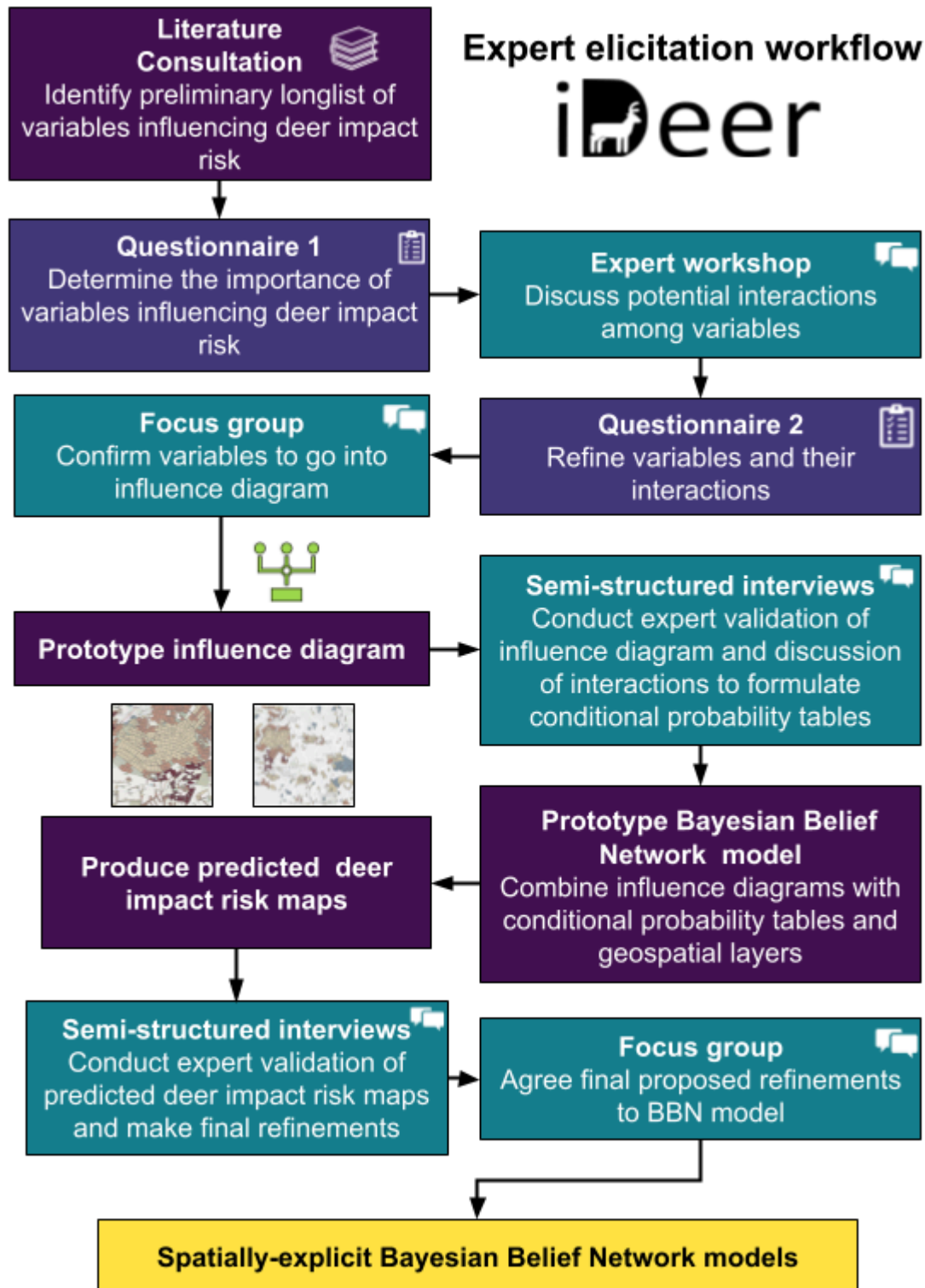
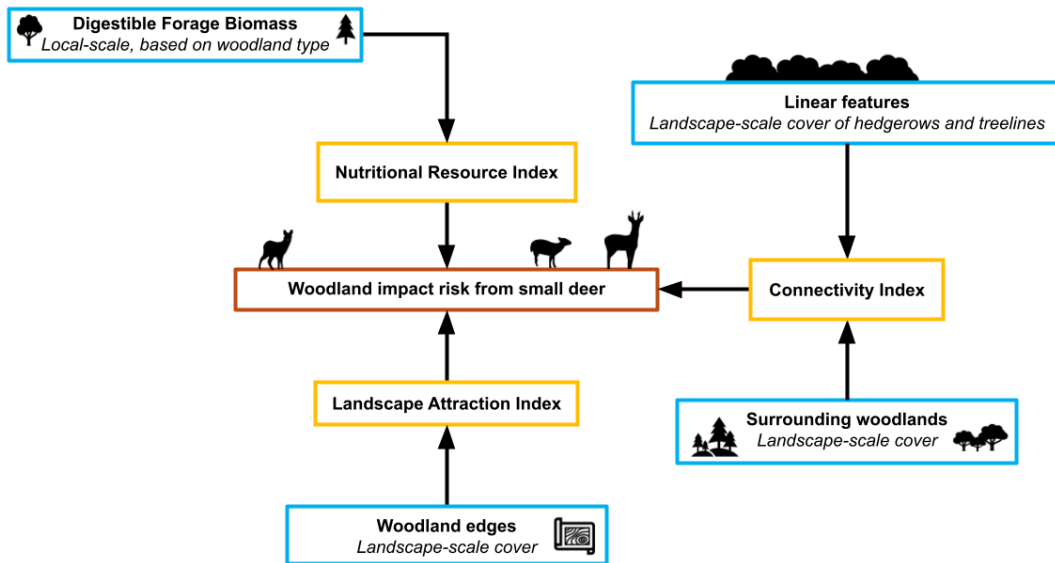


Figure 1: Diagram showing the workflow used to gather information using expert opinion, iteratively acting upon and re-engaging with experts to produce the final models behind the iDeer Tool.

a) Influence diagram for Bayesian Belief Network model predicting impact risk to woodlands from small deer



b) Influence diagram for Bayesian Belief Network model predicting impact risk to woodlands from large deer

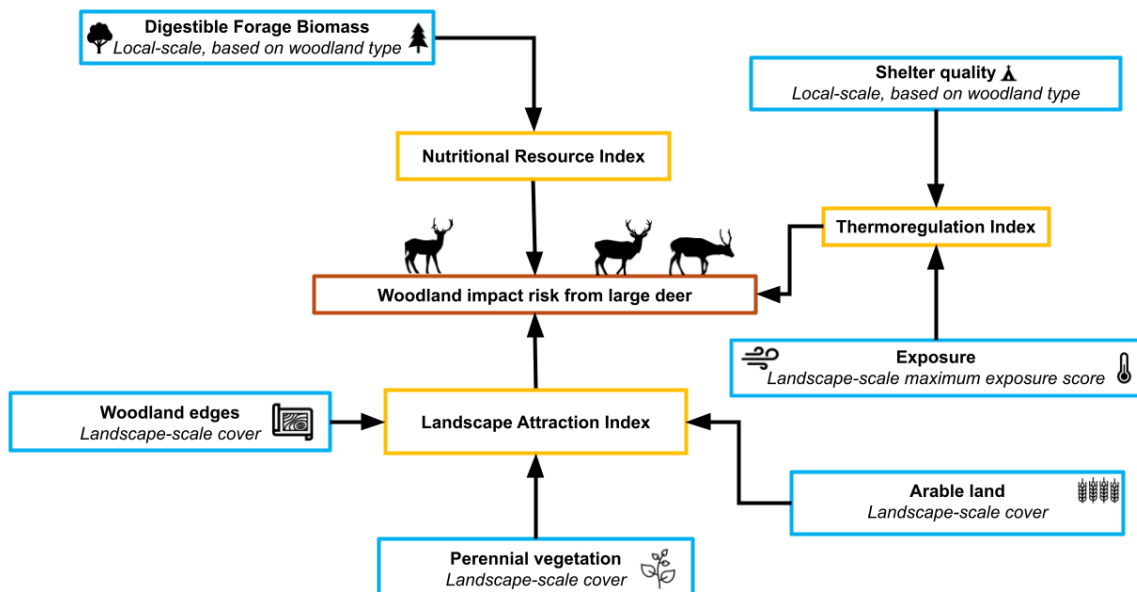


Figure 2: Influence diagrams for the Deer Impact Risk models for a) small bodied deer species (roe, Reeve's muntjac, Chinese water deer) and b) large-bodied deer species (fallow, red, sika). The boxes outlined in blue denote input variables that were measured using publicly available spatial datasets, and the yellow boxes show indices, with Woodland impact risk from large in the orange box in the centre.

How are the variables measured?

The Small and Large Deer Impact Risk models use the variables shown in the blue and yellow boxes in **Figure 2**. The variables in the blue boxes are measured using publicly available and proprietary spatial data. The variables in the yellow boxes represent how deer perceive landscape or woodland attributes, based on a composite measure of its input variables. For example, in the diagram for the Large Deer Impact Risk model, the 'Landscape Attraction Index' indicates the estimated perceived nutritional value of the landscape surrounding an area of woodland, reflecting the positive selection by large deer for landscapes that contain nutritionally rich forage sources (e.g., cropland, productive woodland edges and semi-natural grasslands). The Landscape Attraction Index, along with the two other indices: Nutritional Resource Index (indicating the nutritional value of a local woodland patch) and Thermoregulation Index (indicating the propensity of deer to seek shelter from harsh environments within the woodland), then combine to predict impact risk to each woodland parcel.

Continuous variables (e.g. 0-100%) are converted to a categorical score (low-medium-high) . The thresholds for these scores were determined through expert elicitation and are provided for reference in the Appendix.

What are the key assumptions and limitations of the iDeer Tool?

The iDeer tool is designed to inform deer management decisions, but may not fully represent the actual realised deer damage within woodlands. The maps produced by the tool contain predictions of Deer Impact Risk derived from the models described above. Although we have tried to make our Small Deer Impact Risk and Large Deer Impact Risk models as comprehensive as possible, there are underlying assumptions and limitations that users should be aware of when interpreting deer risk maps produced by the iDeer Tool.

1. The models that the iDeer Tool runs do not incorporate data on deer density

Deer density (e.g., the number of deer per hectare) is not explicitly included in our models. Across England and Wales, deer numbers are highly spatially-variable , fluctuating in response to targeted management efforts, daily activity patterns and resource availability. As a result, spatial data on deer density is highly uncertain.

Instead, we assume 'moderate' deer density across England and Wales. The term "moderate density" is used in this work to describe the deer population to be at a size in the landscape that would cause a measurable impact to vegetation, but not to such an extent that all woodlands have an equal chance of being impacted. We are therefore assuming that deer are likely to exhibit preferential use of certain woodlands over others under moderate densities, which enables the prediction of Deer Impact Risk from local and landscape attributes (i.e., variables shown in blue boxes in Figure 1). Under high densities of deer, it is expected that deer become less 'choosy', in their foraging, and risk of impact can be universally high to all woodlands across a landscape.

2. The models that the iDeer Tool runs do not account for local deer management

Deer management efforts can have substantial local effects on how deer use woodlands across landscapes. However, the models that the iDeer Tool runs do not include information on local deer management. This is because there is no standardised framework for landowners and stalkers across England and Wales to report the type and intensity of their deer management efforts, including

culling, stalking or deer fencing. We note that while culling reduces deer numbers, it does not prevent deer from using woodlands. However, they may become more nocturnal to avoid people.

3. No field data have been used to validate the models that the iDeer Tool runs (yet...!)

The models that the iDeer Tool runs have been validated using an 'expert elicitation and validation' approach. This involved us consulting with a panel of experts familiar with deer behaviour and their management. This expert panel evaluated the credibility of Deer Impact Risk maps produced by the iDeer Tool. Coupling expert opinion with field data would provide more rigorous validation.

4. Deer Impact Risk maps produced by the iDeer Tool complement rather than replace field observations and professional advice

The Deer Impact Risk maps produced by the iDeer Tool are designed to support local- and landscape-level management decisions by highlighting woodlands that are potentially at high risk of deer impacts. However, as described above, the maps use several underlying assumptions. As such, they complement rather than replace on-the-ground assessment of deer activity and impacts and professional advice on deer management.

In England, the [Forestry Commission Deer Management Officer](#) for your region can advise you on how to monitor and manage deer in your woodland. Additionally, NGOs such as the [British Deer Society](#), the [Game & Wildlife Conservation Trust](#) and the [Deer Initiative Partnership](#) also offer deer management guidance for England and Wales.

Who co-developed the iDeer Tool?

Technical development of the iDeer tool was led by Dr Amy Gresham and Dr Becks Spake (University of Southampton) and Dr Matt Guy (Forest Research), with support from Dr Andrew Rattey and Dr Chloe Bellamy (Forest Research).

Development of the Bayesian Belief Network models that underpin the tool was led by Amy Gresham and Becks Spake, supported by the wider project team: Dr Matthew Grainger (Norwegian Institute of Nature Research), Dr Graeme Shannon, Dr Freya St John and Dr Elena Cini (Bangor University), Professor Felix Eigenbrod (University of Southampton), Dr Alastair Ward (University of Leeds), Dr Chris Nichols (Woodland Trust) and Paul Orsi (Sylva Foundation).

Our panel of experts included: David Hooton, Alastair Boston, Jamie Cordery, David Jam, Arman Siddiqui, Andy Page and Maarten Ledebøer (Forestry Commission), Martin Edwards (British Association for Shooting and Conservation), Nick Reed-Beale (Woodland Trust), Adrian Jowitt (Natural England), Nick Jackson (Bowland Deer Management Group & Lune Valley Deer Management Group), Charles Smith Jones (British Deer Society), Dr Robin Gill and Dr Chris Hirst (Forest Research), Dr Owain Barton (University of Alberta), Dr Eilidh Smith (Durham University), Tom Logan (University of Leeds), Dr Ewan McHenry (Woodland Trust), Dr Jochen Langbein (Langbein Wildlife Ecological Consultants) and Professor Rory Putman (University of Glasgow).

More information

We are currently writing an open access scientific journal article. Once published, the full citation and weblink will be provided here. The journal article will provide further technical information on the model building process including the model framework, variable calculation, limitations and

assumptions. The article will also provide a detailed discussion on the use and potential impact of the iDeer Tool.

How to cite the iDeer tool:

Gresham, A., Grainger, M., Guy, M., Bellamy, C., St John, F.A.V., Cini, E., Eigenbrod, F., Rattey, A., Orsi, P., Nichols, C., Logan, T., Lawrence, P., McHenry, E., Hirst, C., Smith, E., Barton, O., Putman, R., Gill, R., Jam, D., Ward, A., Shannon, G. & Spake, B. (2025) The iDeer Tool: A decision-support tool for managing deer alongside woodland creation (Version 1).

How to cite the journal article (in prep):

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Feedback

We would appreciate any feedback on the iDeer Tool design, functionality, guidance and outputs. This feedback will inform future versions. Please go to our [short feedback form](#) to provide anonymous feedback.

Contact us

If you would like to contact us directly with any questions, comments or ideas for future work please email us at ideer.enquiries@gmail.com

Terms and conditions

The iDeer Tool and its contents, ideas and inception was funded as a Future of UK Treescapes project by the Natural Environment Research Council ('iDeer: An Integrated Deer Management Platform', grant no. NE/X003973/1). The information it provides is freely given to support decisions around deer management at the local and landscape scale, not for large businesses, companies or organisations to use to gain profit. It is not to be copied for financial gain. Its information is not to be corrupted in any way. Attribution must be made clearly to iDeer if any parts of the tool or information herein are to be copied or published in any way by anybody.

Appendix: Details of the indices used to predict Deer Impact Risk scores

Who is this section for?

This appendix provides detailed technical information about how the iDeer Tool calculates Deer Impact Risk scores. It's intended for those who wish to understand the underlying methodology. Most users can use the tool effectively without reading this section.

What you'll find here

The following sections explain how we measure and score the different factors that influence Deer Impact Risk. This includes the specific datasets we use, how we categorise woodland types, and the threshold values that determine how a numeric variable is classified into "low," "medium," or "high" categories to go into the Impact Risk models. These categories are based on expert knowledge from deer managers and researchers across England and Wales, combined with scientific literature. The threshold values were determined through workshops and interviews with our expert panel.

Key datasets referenced:

- [NFI \(National Forest Inventory\) 2023](#): Official mapping of all woodlands in Great Britain
- [CEH \(Centre for Ecology & Hydrology\) Land Cover Map 2023](#)
- [Forest Gales 'Detailed Aspect Method of Scoring' \(DAMS\) - Forest Research](#)

Nutritional Resource Index

The Nutritional Resource Index represents a nutritional value of the woodland for deer. In both the large and small deer models, this is calculated from a single variable: **Digestible Forage Biomass**. Digestible Forage Biomass was determined using the ['Interpreted Forest Type' \(IFT\) categories](#). IFT's that are more open and in an early development stage were assumed to have greater Digestible Forage Biomass (the score assigned to each IFT are presented in Table 1). According to our Deer Impact Risk models, woodlands ranked as having higher Digestible Forest Biomass are at greater risk of impact by both small and large-bodied deer species.

Table 1. Digestible Forage Biomass scores for each Interpreted Forest Type in the National Forest Inventory 2023

Interpreted Forest Type	Digestible Forage Biomass for small deer	Digestible Forage Biomass for large deer
Broadleaved	Medium	Medium
Mixed mainly broadleaved	Medium	Medium
Conifer	Low	Medium
Mixed mainly conifer	Low	Medium
Coppice	Medium	Medium
Coppice with standards	Medium	Medium

Assumed woodland	High	High
Failed	High	High
Windblow	High	High
Ground prep	High	High
Felled	High	High
Shrub	High	High
Low density	High	High
Young trees	High	High

Connectivity Index (Small Deer Impact Risk model)

The Connectivity Index is a measure of how well connected a woodland is with other woodlands and woody features in the surrounding landscape. It is only included in the Small Deer Impact Risk Model, as small-bodied deer species tend to remain close to woody cover most of the time. Therefore, a more connected woodland should be more easily accessible to the small-bodied deer species compared to more isolated woodlands. The Connectivity Index is a composite of two variables measured within 400 m of a woodland:

- 1) **Linear features % cover:** The percentage cover of linear features in the surrounding landscape was measured using the UK Centre for Ecology & Hydrology 2016 Woody Linear Features Framework.
- 2) **Woodland % cover:** The percentage cover of woodland in the surrounding landscape was measured using all woodland polygons in the NFI 2023. Expert opinion and validation were used to determine thresholds of low, medium and high woodland percentage cover (see **Table 2**).

Table 2. Thresholds for converting percentage cover of linear features and woodland within the landscape to low, medium and high percentage cover categories for small and large-bodied deer.

Deer body size	Variable	Low	Medium	High
Small deer (roe, muntjac, Chinese water deer)	Percentage cover of linear features within 400m	<=1 %	>1 & <=5 %	>=5 %
	Percentage cover of woodland within 400m	<=2 %	>2 & <=10 %	>=10 %
Large Deer (sika, red, fallow)	Percentage cover of arable within 1000m	0 %	>0 & <=50 %	>50 %
	Percentage cover of perennial within 1000m	<=25 %	>25 & <=50 %	>50 %

Landscape Attraction Index

The Landscape Attraction Index measures the nutritional value of the landscape surrounding the woodland. For the Small Deer Impact Risk model, the Attraction Index was determined by **percentage cover of woodland edges** within a 400m landscape surrounding the woodland (**Table 4**). For the Large Deer Impact Risk model, the Attraction Index is a composite of:

- 1) **Percentage cover of woodland edges within 1000m**, measured using the NFI 2023 (**Table 4**). IFTs (Inferred Forest Types) were categorised into different woodland structures (open or closed woodlands; see **Table 3** for categories) and the boundaries between these two woodland types, as well as between woodland and non-woodland land, were included as woodland edges. See **Table 4** for Low/Medium/High percentage cover categories.
- 2) **Percentage cover of arable land within 1000m**, measured using the UK CEH's 2023 Land Cover Map raster. See **Table 4** for Low/Medium/High percentage cover categories.
- 3) **Percentage cover of perennial vegetation within 1000m** measured using the UK CEH's 2023 Land Cover Map raster. All "Semi-natural grasslands", "Mountain, heath and bog" and "Coastal" land classes that support permanent vegetation were categorised as Perennial land cover. See **Table 4** for Low/Medium/High percentage cover categories.

Table 3. Categorisation of Interpreted Forest Types into structural types

Interpreted Forest Type	Structure type
Broadleaved	Closed
Mixed mainly broadleaved	Closed
Conifer	Closed
Mixed mainly conifer	Closed
Coppice	Closed
Coppice with standards	Closed
Assumed woodland	Open
Failed	Open
Windblow	Open
Ground prep	Open
Felled	Open
Shrub	Open
Low density	Open
Young trees	Open

Table 4. Thresholds for converting percentage cover of woodland edges, arable and perennial land cover into Low, Medium and High categories for the Deer Impact Risk models.

Deer body size	Variable	Low	Medium	High
Small deer	Percentage cover of woodland edges within 400m	$\leq 5\%$	$>5\% \text{ \& } \leq 30\%$	$>30\%$
Large	Percentage cover of woodland edges within 1000m	$\leq 5\%$	$>5\% \text{ \& } \leq 20\%$	$>20\%$
Large	Percentage cover of arable within 1000m	$\leq 0\%$	$>0\% \text{ \& } \leq 50\%$	$>50\%$
Large	Percentage cover of perennial within 1000m	$\leq 25\%$	$>25\% \text{ \& } \leq 50\%$	$>50\%$

Thermoregulation Index (Large Deer Impact Risk model)

The Thermoregulation Index measures how likely large-bodied deer species are to use a woodland for shelter. The Thermoregulation Index is a composite measure of:

- 1) **Exposure** (i.e. the severity of the surrounding environment) Exposure was measured using 'Detailed Aspect Method of Scoring' (DAMS), which indicates the risk of damaging winds to forestry that can also serve as a metric for exposure experienced by animals. We used the maximum DAMS score within 1000-m of a woodland patch to indicate the presence of exposed land within the vicinity of a woodland (see **Table 5**).
- 2) **Shelter Quality** of the woodland itself. Shelter quality was measured using the NFI 2023: Inferred Forest Types (IFTs) were classified into low, medium and high shelter quality categories according to canopy openness and development stage (see **Table 6**).

Table 5. Thresholds for converting maximum DAMS scores within a 1000m landscape to low, medium and high scores for the Exposure variable in the Large Deer Impact Risk model. Note that DAMS scores do not have units.

Variable	Low	Medium	High
Maximum summed DAMS within 1000m (normalised from 0-1)	≤ 0.6	$> 0.6 \text{ \& } \leq 0.80$	> 0.80

Table 6. Shelter quality score used for each Interpreted Forest Type.

Interpreted Forest Type	Shelter quality
Broadleaved	High
Mixed mainly broadleaved	High
Conifer	High
Mixed mainly conifer	High
Coppice	Medium
Coppice with standards	Medium
Assumed woodland	Low
Failed	Low
Windblow	Low
Ground prep	Low
Felled	Low
Shrub	Medium
Low density	Low
Young trees	Low

